

Maximum Nullity of Cyclic Covers of Graphs: A Monodromy Ceiling and the Zero Forcing Number of Generalized Petersen Graphs

Category: Mathematics

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How degenerate can the zero energy level of a periodic operator on a graph be? We answer it **exactly and unconditionally** for infinite families of covers — proving a conjecture posed in 2026, and correcting a published theorem that is false.

INTRODUCTION — the question

The problem. Over all symmetric matrices with off-diagonal pattern G — Hamiltonians on that network — how degenerate can the zero eigenvalue be? That maximum $M(G)$ is the central quantity of the **inverse eigenvalue problem of a graph**.

The tool. $M(G) \leq Z(G)$, the **zero forcing number**, a colouring game from quantum control and power-grid monitoring.

The family. $P(n, k)$ is a **cyclic cover**: one base graph repeated n times with a twist. This is the spectral theory of a periodic structure, so any answer must hold *uniformly in the period n* .

What was wrong. The published value is **false**: Rashidi et al. (2020) claim $Z(P(n, 3)) = 8$ for $n \geq 12$, but $Z(P(12, 3)) = 7$.

INTRODUCTION — the gap

The 2026 note states as **Conjecture 5** that $Z(P(n, 3)) = 8$ for $n \geq 13$, and says what is missing:

“the content is the lower bound $Z(P(n, 3)) \geq 8$ for all $n \geq 13$, which we verified only for $n \leq 20 \dots$ We will not assert it from finite computation, which would reproduce the gap this note corrects.”

It closes: *“Determining $Z(P(n, k))$ for general $k \dots$ remains open.”*

So the gap is uniformity in n , not more computation:

- ▶ combinatorial bounds must exclude all subsets at once;
- ▶ matrix bounds need one matrix per n — infinitely many.

Prior work treats circulants; $P(n, k)$ is a *block* circulant, so it is not covered.

FRAMEWORK

The graph. $P(n, k)$: outer u_i , inner v_i , edges $u_i u_{i+1}$, $u_i v_i$, $v_i v_{i+k} \pmod n$. Cubic, and $\rho(i) = i + 1$ is an automorphism — a **cyclic cover** of a two-vertex voltage graph.

The matrix class. $\mathcal{S}(G) = \{A = A^\top : a_{ij} \neq 0 \iff ij \in E, i \neq j\}$, and $M(G) = \max \text{null } A \text{ over it.}$

Two facts. $M(G) \leq Z(G)$ and $Z(P(n, k)) \leq 2k + 2$.

Goal. Exhibit $A \in \mathcal{S}(P(n, k))$ with $\text{null } A = 2k + 2$. Then $2k + 2 \leq M \leq Z \leq 2k + 2$, so $Z = M = 2k + 2$ — an exact value, not a bound.

FINDINGS 1 — an exact ceiling

Theorem 1 (reduction), $n \geq 2k + 3$

Eliminating the inner coordinates turns $Ax = 0$ into one linear recurrence of order **exactly** $2k + 2$, with monodromy T , and $\text{null } A = \dim \ker(T - I) \leq 2k + 2$, equality iff $T = I$.

Proof sketch. The outer row meets *one* inner variable, whose spoke weight is nowhere zero. Dividing it out leaves a nine-term relation with extreme offsets $\pm(k + 1)$, nonzero.

The ceiling equals the target. $2k + 2$ is also the best known *upper* bound on Z : the method **either settles Z or cannot touch it**. The proof never used the outer voltage, so it holds on **every** two-vertex cover, ceiling $2(p + q)$ — this is $p = 1$.

FINDINGS 2 — an obstruction

Singularity is governed by a **symbol** $F(s) = (s + a)L_k(s) + \alpha s + \beta$, monic of degree $k + 1$ in $s = 2 \cos(2\pi m/n)$, with **exactly three** free parameters for every k : three roots may be prescribed, no more. Realisability is $c^2 = a\alpha - \beta$.

Theorem 2 (antipodal obstruction)

$c^2 = 0$ identically $\iff F = (s + a)(L_k + t)$. Since $L_k(-s) = (-1)^k L_k(s)$, for **every even** k a symbol with roots $y, -y$ has $c^2 = 0$; for odd k there is no such obstruction.

Proof sketch. c^2 vanishes identically exactly when F factors through L_k , and the parity identity turns an antipodal pair into that factorisation.

FINDINGS 3 — the congruence wall

A rational symbol drags its whole Galois orbit onto the grid. Taking F as a product of minimal polynomials of $2 \cos(2\pi/d)$ gives **exact integer** certificates: $Z(P(n, 4)) = 10$ for $60, 70, 90 \mid n$ — the first exact values for $k > 4$.

Theorem 3 (the congruence wall)

Every rotation-invariant certificate is confined to **one** divisibility class, containing **at most one prime**; and $e_2 = -k$ for every such symbol with $k \geq 3$.

Proof sketch. Grid roots must form full Galois orbits, forcing $\text{lcm}(D) \mid n$. So symmetry is not a shortcut but a **barrier**: to reach all n it must be broken.

FINDINGS 4 — tiling

Drop rotation invariance: $T = T_{n-1} \cdots T_0$, and we need $T = I$.

Docking. T_i sees only $[i - k - 1, i + k]$, so fixing $k + 1$ weights at each end of a tile makes its product independent of its neighbours: tiles **compose**.

Theorem 4 (tiling)

If an identity tile exists for every length in $[L, 2L)$, then $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for **every** $n \geq L$.

Proof sketch. Every $n \geq L$ is a sum of lengths from $[L, 2L)$; concatenating gives $T = I$. In fact $[L, 2L)$ is only *sufficient*: n need only be a *sum* of certified lengths.

An all- n theorem, reduced to finitely many finite problems.

FINDINGS 5 — exact certification

$T_{\text{tile}} = I$ is rational in the weights, so bounding it directly needs interval matrix products. Avoided: by Theorem 1,

$T_{\text{tile}} = I \iff \text{null } A_{\text{one tile}} = 2k + 2$, certified by the **bilinear** system $A(w)K(X) = 0$, whose affine Jacobian lets a **Krawczyk contraction** need only three norms.

And the arithmetic is exact. The box centre is dyadic, f and J have integer coefficients, and Y is *arbitrary* — so rounding it to a rational costs nothing. Balanced limb decomposition keeps the product exact at BLAS speed.

All **77** tiles pass, $\alpha \leq 1.0 \times 10^{-9}$; each $\alpha < 1$ is an **exact integer comparison**, not an estimate.

FINDINGS 6 — results

k	settled	L	certified tiles	$\max \alpha$
2	every n	—	closed form	—
3	every n	21	23	3.2×10^{-10}
4	every n	29	33	7.6×10^{-10}
5	every $n \geq 162$	54	21	1.0×10^{-9}

- ▶ $Z(P(n, 3)) = 8$ for every $n \geq 13$, and 13 is optimal — this **proves Conjecture 5**.
- ▶ $Z(P(n, 2)), Z(P(n, 3)), Z(P(n, 4))$ determined for **every** n , the last two for the first time. **No divisibility conditions; primes included.**

The $k = 3$ sequence runs 6, 6, 6, 8, 7, 7, 8, 8, ... — non-monotone, which is how the published proof went wrong.

CONCLUSIONS

How this answers the question. The ceiling makes the method exactly as strong as the target; the congruence wall shows symmetry cannot reach all n ; tiling breaks it and supplies the **uniformity in n** the 2026 note identified as missing.

Limitations.

- ▶ $k \geq 6$: no tiles. $k = 5$: threshold 162 not optimal.
- ▶ $M(P(n, 4))$ open at one value, $n = 24$: there $Z = 10$, no certificate.
- ▶ Whether $M = Z$ in general is open; the rank law is verified, not proved.

Unexpected problems. A stalling search was twice mistaken for non-existence — which is why every search now runs with a control.

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